ChatGPT edition — read and model only; produce no sparks

Lesson 10: Spark Gaps

Gas breakdown as historical physics, not a learner build

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

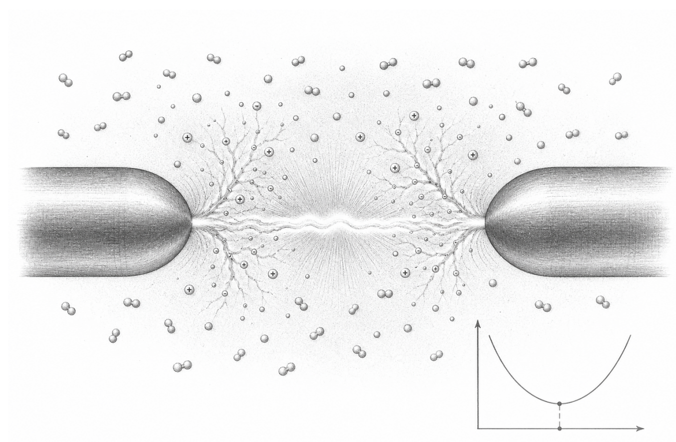
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- This reading
- Printed Paschen-curve sketch
- Pencil
- No energized apparatus

Predict first

Why is "air is an insulator" incomplete rather than simply false?



Conceptual cutaway, not construction guidance: collisions multiply charge carriers until a conductive channel forms. The inset minimum belongs to a pressure–gap curve, not to gap alone.

Investigate

1. Trace the historical sequence: charge separation, increasing field, ionization, conductive channel, rapid energy transfer, recovery.
2. On the curve, mark why breakdown is not determined by gap alone.
3. Identify hazards omitted by a pretty spark photograph: shock path, burns, fire, ultraviolet light, sound, ozone, and radio interference.
4. Compare this mechanism with the low-voltage switch pulses in Lessons 6 and 9.

Explain from the evidence

A gas can become conductive when the field and collision conditions sustain ionization. Breakdown depends on gas, pressure, geometry, and separation. A spark gap is a nonlinear switch, historically useful in oscillators but noisy and hazardous. Nothing in this lesson authorizes construction: exposed high voltage remains qualified-person-only.

Change one thing

Write a museum-label explanation of a historical spark transmitter that includes both energy flow and why it is unsuitable for learner replication.

Evidence record	
Prediction	________________
One change	________________
Observation	________________
Claim + because	________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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